Unit 2 Quiz B

Forces & Translational Dynamics · AP-style multiple choice

12 questions + 1 optional Time: **24 min** (AP pace ≈ 2 min each) Calculator OK g = 9.8 m/s²

- Before choosing, **write one line of work** in the margin: an FBD sketch, an equation, or a ratio. Those are the habits the FRQs grade.
- Circle your confidence for every question. A lucky guess still counts as a gap.
- Grade with the coaching guide. For every miss AND every 'guess', read the full walkthrough and do the 'Go deeper' question.

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1. `2.1 Center of mass (2-D)`

Three particles: 1.0 kg at (0, 0), 1.0 kg at (4.0 m, 0), and 2.0 kg at (0, 3.0 m). Where is the center of mass?

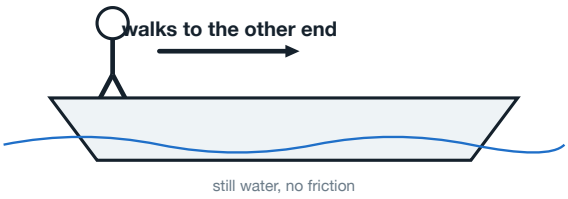
A (1.0 m, 1.5 m) **B** (1.33 m, 1.0 m) **C** (2.0 m, 1.5 m) **D** (1.0 m, 1.0 m)

Confidence: sure | 50/50 | guess Answer: ____

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2. `2.1 Systems`

A person stands at one end of a boat floating at rest on still water (ignore water friction). The person walks to the other end. What happens?



A The boat stays still, because the person's forces are internal

B The center of mass of the person + boat system moves along with the person

C The system's center of mass stays put, and the boat slides backward

D The boat moves in the same direction as the person

Confidence: sure | 50/50 | guess Answer: ____

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3. `2.2 Free-body diagrams`

A ball is thrown at an angle. At the HIGHEST point of its path (air resistance negligible), which forces act on it?

A Gravity down, and a horizontal force that keeps it moving forward

B Only gravity, downward

C No forces, since it is momentarily at the top

D Gravity down and an upward force from the throw that has just run out

Confidence: sure | 50/50 | guess Answer: ____

4. `2.3 Newton's 3rd law`

A heavy truck pushes a small car with a dead battery. They speed up together. How does the force of the truck on the car compare with the force of the car on the truck?

A Truck on car is greater, otherwise they couldn't speed up

B They are equal only once they reach a constant speed

C They are equal at all times

D Car on truck is greater, because the car resists

Confidence: sure 50/50 guess Answer: ____

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5. `2.4 / 2.5 Net force`

A skydiver opens her parachute and slows from 50 m/s to 6 m/s while still falling. During the slowing, how does the air-drag force compare with her weight?

A Drag is less than her weight, since she is still moving down

B Drag equals her weight

C Drag is greater than her weight

D Drag is zero once she slows down

Confidence: sure 50/50 guess Answer: ____

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6. `2.5 Factor of change`

A net force F gives an object of mass m an acceleration a. What acceleration does a net force 2F give an object of mass 3m?

A a/3 **B** 2a/3 **C** 6a **D** 3a/2

Confidence: sure 50/50 guess Answer: ____

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7. `2.6 Universal gravitation`

Two spheres attract each other with force F. The mass of ONE sphere is doubled and the distance between their centers is tripled. The new force is:

A 2F/3 **B** 2F/9 **C** 6F **D** 4F/9

Confidence: sure 50/50 guess Answer: ____

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8. `2.6 Apparent weight`

An elevator's cable snaps and it falls freely (ignore air resistance). A rider stands on a scale inside. During the fall:

A Gravity on the rider becomes zero, so the scale reads zero

B The scale reads zero, but the gravitational force on the rider is still mg

C The scale still reads mg

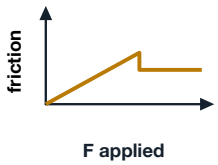
D The scale reads 2mg

Confidence: sure 50/50 guess Answer: ____

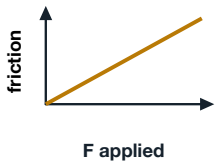
9. `2.7 Static vs kinetic`

A box sits on a rough floor. A horizontal push starts at zero and increases slowly until well after the box starts to slide. Which graph best shows the friction force vs. the applied force?

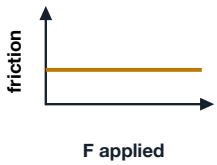
A



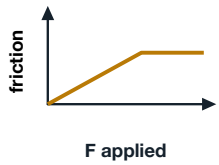
B



C



D



Confidence: sure 50/50 guess Answer: ____

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10. `2.7 Incline + kinetic friction`

A block slides DOWN a ramp of angle θ at constant speed. The coefficient of kinetic friction must be:

A sin θ **B** cos θ **C** tan θ **D** 1 / tan θ

Confidence: sure 50/50 guess Answer: ____

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11. `2.9 Flat curve`

A car rounds a flat, unbanked curve of radius 50 m. The coefficient of static friction between the tires and road is 0.80. What is the maximum speed without skidding?

A 19.8 m/s **B** 14.0 m/s **C** 6.3 m/s **D** 392 m/s

Confidence: sure 50/50 guess Answer: ____

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12. `2.9 Orbits · BOSS`

A satellite in a circular orbit of radius r has speed v and period T. It is moved to a circular orbit of radius 4r around the same planet. Its new speed and period are:

A v/2 and 8T **B** v/4 and 4T

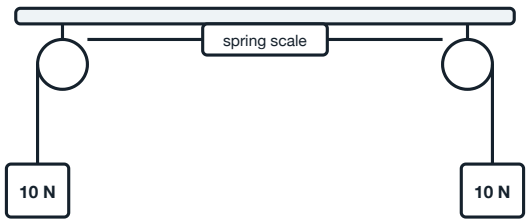
C 2v and T/8 **D** v/2 and 4T

Confidence: sure 50/50 guess Answer: ____

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13. `2.8 Springs · optional: ask if springs are on the test`

A spring scale is connected between two strings. Each string runs over a pulley and holds a 10 N weight, so the system is at rest. What does the scale read?



A 0 N **B** 10 N **C** 20 N **D** 5 N

Confidence: sure 50/50 guess Answer: ____